

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA070 2

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I Deepak-19241021 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Deepak

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Subnetting a /24 Network into Multiple Subnets.

Given,

- Network Address: 192.168.20.0/24
- Lab A $\rightarrow$ /26
- Lab B $\rightarrow$ /27
- Lab C $\rightarrow$ /28

Procedure:

Subnetting divides a large network into smaller subnets to improve network management.

• Lab A (/26):

- $\rightarrow$ Network Address: 192.168.20.0
- $\rightarrow$ Host Range: 192.168.20.1 - 192.168.20.62
- $\rightarrow$ Broadcast Address: 192.168.20.63
- $\rightarrow$ Usable Hosts: 62

• Lab B (/27):

- $\rightarrow$ Network Address: 192.168.20.64
- $\rightarrow$ Host Range: 192.168.20.65 - 192.168.20.94
- $\rightarrow$ Broadcast Address: 192.168.20.95
- $\rightarrow$ Usable Hosts: 30

Lab C (28):

→ Network Address: 192.168.20.96

→ Valid Host Range: 192.168.20.97 -

→ Broadcast Address: 192.168.20.111

→ Usable Hosts: 14

Broad
IPA

2) Subnet Analysis for Two Systems using /26 Subnet Mask.

Given,

System 1: 192.168.1.130/26

System 2: 192.168.1.190/26

Subnet Mask: 255.255.255.192

Solution,

System 1 (192.168.1.130/26):

• Network Address: 192.168.1.128

• Valid Host Range: 192.168.1.129 - 192.168.1.190

• Broadcast Address: 192.168.1.191

• IP Address Status: Valid

System 2 (192.168.1.190/26):

• Network Address: 192.168.1.128

• Valid Host Range: 192.168.1.129 - 192.168.1.190

Broadcast Address: 192.168.1.128

IP Address Status: Valid

3) Subnet Allocation for Organizational Departments

Given,

Network Address: 172.16.0.0/16

• ~~HR~~ → 125

• HR → 125, IP → 127

Sales → 126, Admin → 128

Sol:

HR (125):

• Network Address: 172.16.0.0 → 17

• Host Range: 172.16.0.1 - 172.16.0.126

• Broadcast Address: 172.16.0.127

• Usable Hosts: 126

Sales (126):

• Network Address: 172.16.0.128

• Host Range: 172.16.0.129 - 172.16.0.190

• Broadcast Address: 172.16.0.191

• Usable Hosts: 62

IT (127):

- Network Address: 172.16.0.192
- Valid Host Range: 172.16.0.193 - 172.16.0.223
- Broadcast Address: 172.16.0.223
- Usable Hosts: 30

Admin (128):

- Network Address: 172.16.0.224
- Host Range: 172.16.0.225 - 172.16.0.238
- Broadcast Address: 172.16.0.239
- Usable Hosts: 14

Remaining Unused IP Range:

- 172.16.0.240 - 172.16.0.255

4) Subnet Identification Using a /27 Subnet Mask:

Given,

- Network Address: 172.16.10.0/24
- Subnet Mask (127): (255.255.255.224)
- IP address: 172.16.10.78
- Check IP: 172.16.10.95

sol,

Total Subnets: 8 subnets

Usable Hosts per Subnet: 30 Hosts

Subnet for 172.16.10.78

- Network Address: 172.16.10.64
- Host Range: 172.16.10.65 - 172.16.10.94

Verification of 172.16.10.92

- 172.16.10.95 is the Broadcast Address
- It is not a valid Host address and cannot be assigned to a device.

5) IPv6 Address Compression and Network Prefix Calculation

Given,

IPv6 Address: 2001:0db8:0000:0000:0000:0000:
0042:8329

Prefix Length: /64

Solution,

Compressed IPv6 Address:

→ 2001:db8::ff00:42:8329

Expanded IPv6 Address:

→ 2001:0db8:0000:0000:0000:ff00:0042:8329

Network Prefix (/64):

→ 2001:db8:0:0::/64

Total Possible Host Addresses:

Q. → $2^{64} = 18,446,744,073,709,551,616$